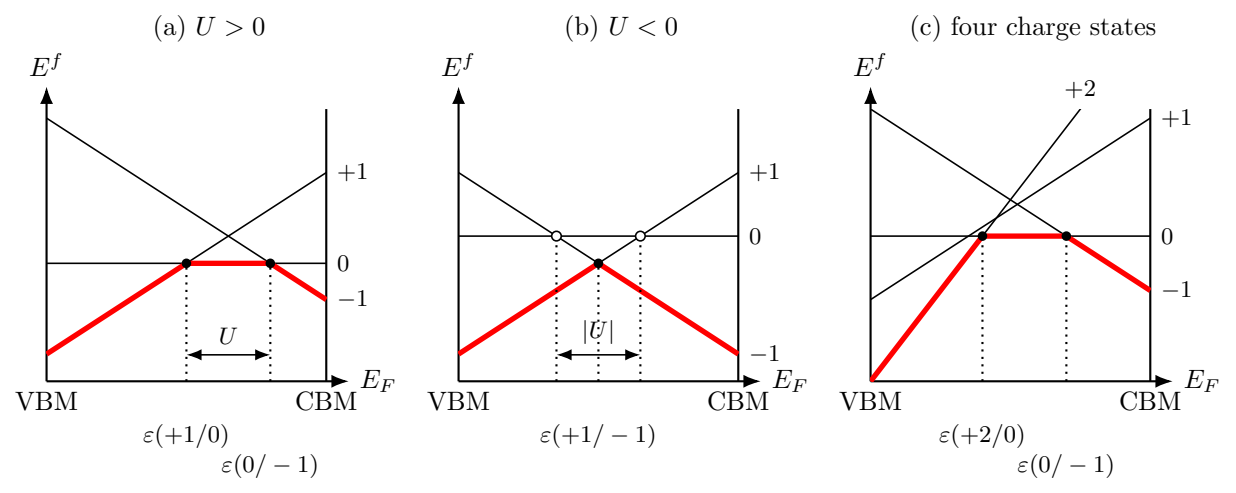

(a) $U > 0$
(b) $U < 0$
(c) four charge states
E^f
+2
+1
0
−1
U
$|U|$
E_F
VBM
CBM
$\varepsilon(+1/0)$
$\varepsilon(0/-1)$
$\varepsilon(+1/-1)$
$\varepsilon(+2/0)$
$\varepsilon(0/-1)$